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Optimal substation capacity planning method in high-density load areas considering renewable energy

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Abstract

With the increasing penetration of renewable energy, the adaptability of the existing substation planning model in terms of capacity and quantity of transformer needs to be further studied when preferring large-capacity substations. Considering the variations of renewable energy penetration rate and load, this paper proposes a method to optimize the total capacity of substations in distribution networks. This paper introduces the influence of renewable energy access on power supply reliability and introduces the idea of partitioning for economic analysis of reliability. An economic analysis model for simultaneously optimizing the capacity and quantity of substation transformer in the distribution network is constructed, taking into account the effects of reducing net load and enhancing the reliability of the distribution feeders resulting from renewable energy access to the medium and low voltage side of the substation. Various wiring means of the distribution network are retained and the impact of renewable energy access on the reliability of the network power supply is quantified. The optimization model is solved by the multivariate universe optimizer(MVO) algorithm with stronger optimality finding capability and short solving time. Finally, the case study results of a regional distribution network are employed to demonstrate and verify the validity and rationality of the method.

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Keywords: Large capacity substation; Renewable energy; Multivariate universe optimizer; Reliability; Economic analysis

1. Introduction

With the rapid development of modern society, the urban load density increases continuously, resulting in transformers in frequently heavy-loaded condition and more demand for substations and land. For instance, the maximum electricity load of the Guangdong power grid, as the load center of the southern region of China, is

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expected to be 173,000 MW in 2025 and there will be 2365 substations of 110 kV voltage level with a capacity of 252,140 MVA. In order to solve the contradiction between grid substations construction and land shortage in developed areas, it is necessary to allocate larger capacity substations for loads, so as to reduce the resources occupation of stations and lines and ensure that the planning schemes can be implemented in time.

Large capacity substations in terms of capacity and quantity of transformer can increase the power supply capability of a closed area, but also require a greater number of incoming and outgoing lines from different voltage levels. Further, the capacity and scale of main transformers can perform well in need of the match of the number of wiring and proper wiring types, which are related to the economy and reliability from the perspective of network interconnection strength. Therefore, it is necessary to consider substation scale impact on different operation and management indicators, such as load rate, loss and reliability, etc. Ref. [1] points out that different wiring structures have different effects on power supply reliability, in terms of ring rate and component failure rate. Ref. [2] indicates that the scale of urban distribution systems is growing, and additional losses may occur due to improper grid structures.

Clearly, the increasing penetration of renewable energy affects the selection of substation capacity for the net load to be put down. However, the uncertain generating way of renewable energy featured by salient randomness cannot exert its full capacity just as its rated power. In addition, the renewable energy can also bring positive effects to the reliability improvement of the distribution network, and appropriately reduce the requirements for substation capacity under the maintenance of power supply reliability targets [3,4]. When a fault occurs on some middle point of the section line, the fault and the affected portion of the line are disconnected from the substation, while the end part of the line in a single source wire structure must fail. But such power failure can be reduced or even avoided when renewable energy sources access parts of the distribution network. Ref. [5] proposes that distributed photovoltaic integrated to the distribution network can improve the reliability of the original system. With the increase of distributed photovoltaic capacity, all of the average outage time, the frequency of outages and the average outage power of the system decrease gradually. Therefore, it is necessary for system planners to study the coordination between substation capacity optimization and renewable energy consumption in high-density load areas.

Existing substation planning models can be divided into two categories. The first category mainly focused on the optimal locations and capacity of the substation. Ref. [6] presents a new model for the optimal locations and capacity of substations based on the whole life cycle cost of equipment. It involves the one-time investment cost of the substation, the operation and maintenance cost of the system, the failure cost and the disposal cost. Ref. [7] considers the minimum investment and annual operating cost as the objective function to determine the location of the substation, the transformer capacity combination, and the power supply range. The second category mainly focuses on the optimal power supply radius. In Ref. [8], a substation radius optimization model was constructed to find the minimum annual cost per unit of the supply area. Ref. [9] considers the capacity–load ratio constraint applicable to large capacity transformers. It investigates the relationship between transformer capacity, transformer number, and supply radius, and gives its optimal combination. Ref. [10] compares the indicators of comprehensive investment, operating costs, reliability of power supply, and conditions for the construction of the grid, for electricity demanded by regional development. In this way, the transformers' number, capacity and power supply radius are defined, and the number and length of lines are estimated.

Based on the references, existing substation planning models mainly focus on the relationship between three variables: load density, supply radius, and substation capacity. They usually study two of three variables by fixing the remaining one for simplicity. As for some potentially larger substation schemes, such as $4 * 63$ MVA and $3 * 80$ MVA in 110 kV system, few studies have been compared in a comprehensive model, considering their similar total substation capacity but the unequal number of transformer units and discriminative individual capacity. In addition, the existing models rarely consider the impact of renewable energy access and involve the reliability cost analysis under different grid wiring means [11,12]. The adaptability of this traditional substation planning model might weaken when the penetration of renewable energy sources increases.

Thus, this paper presents an optimization model of the capacity and number of substation transformers in the distribution network with minimum construction and operation costs as the objective function. It is characterized by the reliability cost of renewable energy access and the grid structure of the transformer's secondary side. The impact of renewable energy access on feeders reliability in terms of network structures is considered in the proposed model, which makes the economic indexes of reliability more important and subtler in a whole problem. The reliability index is calculated by the idea of partitioning for the sake of reliability estimation efficiency. Based on the features of the model, an algorithm called the multivariate universe optimizer (MVO) is made to solve the problem. Some examples are tested and the results of the case study verify the superiority of the method in substation planning, which would be a promising tool for the selection of the number of transformers in large capacity substations.

The rest of this paper is organized as follows. The Section 2 analyzes the impact of renewable energy access on power supply reliability. Section 3 presents the economic analysis model for substation construction and operation. Section 4 introduces the MVO algorithm. Section 5 shows the case results of a regional distribution network in the Guangdong power grid. Finally, the main conclusions of this study are summarized in Section 6.

2. Influence of renewable energy access on power supply reliability

2.1. Economic analysis of reliability of distribution network connection mode

Reliability is one of the important evaluation indexes of distribution network connection mode planning, which mainly reflects the continuous power supply capability of the power system. The economic analysis of the reliability of the distribution connection mode, or network structure, in this paper is to convert the reliability index into economic cost on the premise of meeting the requirements of power supply reliability, and its core is the calculation of the economic cost of reliability. The main evaluation index of distribution reliability is the system average power supply reliability rate [13].

$$Rs = \left[1 - \sum_{i=1}^{m}(N_i\mu_i)/\sum_{i=1}^{m}(8760N_i)\right] \times 100\% \tag{1}$$

where m is the number of loads in the typical wiring of the distribution network; N_i is the number of customers at load i, and μ_i is the average annual outage time at the load point i.

Different wiring methods will correspond to different reliability economic costs. Referring to the literature [14], the reliability indexes and the reliability economic cost per unit load for typical wiring methods of urban medium voltage distribution networks obtained under a unified distribution network planning scenario are shown in Table 1.

Table 1. Reliability index and reliable economic cost per unit load for each typical wiring method of urban medium voltage distribution network.

Typical wiring methods	Reliability index Rs	Economic index/(million yuan/MW)			
		Construction cost	Operating cost	Outage loss cost	The economic cost of reliability
Double-loop network	99.99865%	35.1750	11.5567	5.664	52.3957
Single-loop network	99.99804%	27.1455	9.6644	8.256	45.0659
2-contact and 3-section network	99.99136%	29.8220	10.4104	36.288	76.5204

As can be seen from Table 1, without considering renewable energy access, the research results for typical wiring show that the reliability of double-loop network, single-loop network and 2-contact and 3-section network is close, but the economic cost of reliability varies greatly. In terms of outage loss costs, the 2-contact and 3-section network is more than six times higher than the double-loop network. Therefore, the planning of large-capacity substations must take into account the impact of different wiring methods.

2.2. Economic analysis of reliability considering renewable energy access

Distributed generator is connected to the distribution network as planned, which makes a fact that there might be more than one power source in a feeder. When a fault occurs in the system, after fault isolation and switch switching, different areas on a feeder have different operation modes according to the power source location. These areas can be roughly divided into four categories [15]: Class A partition: fault occurrence area, only when the fault is repaired, this partition can resume power supply; Class B partition: fault affected area, the outage time of such partition depends on the sum of fault isolation time and switch switching time; Class C partition: islanded area with the distributed generator, the outage time of this partition depends on the fault isolation time; Class D partition: fault-free area, this partition is located upstream of the fault point, close to the substation.

Taking the radial network as an example, as in Fig. 1, the system is divided into seven zones based on switching elements, where zone Z_4 is connected to the renewable energy DG. It is assumed that only the switch between Z_1 and Z_2 is a circuit breaker and has an automatic opening and closing function, while the others are non-automatic load switches. When a fault occurs at point K in zone Z_3, the seven zones can be further divided into 4 class sets, which are $A = \{Z_3, Z_6, Z_7\}$, $B = \{Z_2\}$, $C = \{Z_4, Z_5\}$, and $D = \{Z_1\}$. It can be seen that, due to the addition of DG, two regions in C can restore power supply, making the average partition outage time lower.

It is known that the outage loss in class D partition is 0, while the outage loss in class A\B\C partitions will be affected by the outage time and outage load. Therefore, typical faults in all areas are counted and the areas are classified, and thus the average outage time in a year is calculated for class $A\backslash B\backslash C$ partitions.

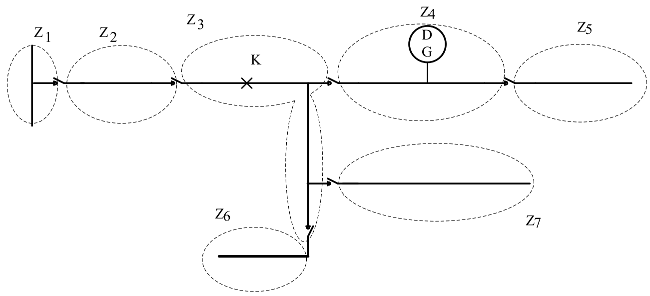

Fig. 1. Zoning map of the impact of renewable energy access.

(1) Class A partition: fault occurrence area

$$T_A = \sum_{i=N_A} \lambda_i + \gamma_i + \sum_{i=N_A} \lambda'_i \gamma'_i \tag{2}$$

$$L_A = P_A \times T_A \tag{3}$$

where λ_i and λ'_i are the failure rate and planned maintenance rate of the components, respectively; γ_i and γ'_i are the average outage duration and average planned outage duration of the components, respectively; N_A is the number of all components in the class A partition; L_A is the average annual outage loss; and P_A is the average annual outage load.

(2) Class B partition: fault affected area

$$T_B = \sum_{i=N_B} p_i \sum_{j=N_B} \lambda_j t_1 + \sum_{i=N_B} (1 - p_i) \sum_{j=N_B} \lambda_j t_2 + \sum_{j=N_B} \lambda'_j \gamma'_j \tag{4}$$

$$L_B = P_B \times T_B \tag{5}$$

where t_1 is the fault isolation and switching time; t_2 is the fault repair time; p_i is the probability that component i in the class B partition can be supplied with distributed electricity; N_B is the number of all components in the class B partition. L_B is the average annual outage loss; P_B is the average annual outage load.

(3) Class C partition: island formation area

$$T_C = \sum_{i=N_C} p_i \sum_{j=N_C} \lambda_j t_3 + \sum_{i=N_C} (1 - p_i) \sum_{j=N_C} \lambda_j t_2 + \sum_{j=N_C} \lambda'_j \gamma'_j \tag{6}$$

$$L_C = P_C \times T_C \tag{7}$$

where t_1 is the fault isolation time; N_C is the number of all components in the class C partition. L_C is the average annual outage loss; and P_C is the average annual outage load.

(4) Class D partition: without impact area

The average outage rate and average outage time within such areas are both zero.

After a fault occurs, through fault isolation and switchover, an island is formed in part of the partition, and the probability of reliable power supply from distributed power sources can be obtained by determining the loads within the island. The average outage load and average outage time for each type of partition with different fault conditions are summed up. Thus, the reliability of power supply by renewable energy access in a region can be calculated from Eqs. (2)–(7).

3. Optimization model

A substation optimization model is presented in this paper to consider the economy of large-capacity substation construction and system operation, like the reliability of secondary distribution network, with renewable energy involved. By quantifying the reliability cost and the influence of renewable energy on the substation scale directly and indirectly, the optimization model to minimize the total cost of 10 years of construction and operation is obtained. The decisive variables are the number of transformers, the capacity of each transformer, and the number of incoming and outgoing lines of the primary and secondary sides of the transformer. The model not only considers the characteristics of the substation itself and its subordinate distribution network, but also takes into account various connection modes of the distribution feeders and analyzes the impact of renewable energy access on the network power supply reliability. In general, the total construction and operation cost of substations include five parts: input line investment cost (f_1), outgoing line investment cost (f_2), substation investment cost (f_3), substation operation and maintenance cost (f_4), and reliability cost (f_5).

Take the 110 kV substation with the low-voltage side of 10 kV as an example to build the following optimization model. The optimization model is based on the following assumptions:

- The power supply area is circular and the substation is located at the ideal center.
- The annual growth rate of load density is the same every year.
- Transformers of equal capacity have the same number of outgoing lines.

3.1. Objective function

The objective function expression of the optimization model is as follows:

$$\min F = \sum_{t=1}^{n} f(t) \times (1+k)^{n-t} \tag{8}$$

where *f(t)* is the total construction and operation costs of the substation in tth year; k is the discount rate of the electric power industry; n is the economic operation life of the equipment.

It is assumed that the supply radius of the large capacity substation is R, and the capacity of transformer is S_i and *N(t)* is the number of transformers in the substation in tth year. Then the construction and operation cost of the substation in the ith year can be obtained as follows:

$$f(i) = f_1 + f_2 + f_3 + f_4 + f_5 \tag{9}$$

- Investment cost of incoming lines

The investment cost of the incoming lines can be expressed as follows.

$$f_1 = (C_0 + \alpha R C_1) \times [M_1(t) - M_1(t-1)] \times N(t) \tag{10}$$

where C_0 is the constant cost of each incoming line, C_1 is the investment cost of a unit length of the incoming line, $M_1(t)$ is the number of incoming lines in the year t. α is the incoming line zigzag coefficient, and its value is taken as 2[9].

- Investment cost of outgoing lines

$$f_2 = (C_2 + R C_3) \times [M_2(t) - M_2(t-1)] \times N(t) \tag{11}$$

where C_2 is the constant cost of each outgoing line, C_3 is the investment cost of a unit length of the out-line, $M_2(t)$ is the number of outgoing lines in year t.

- Investment cost of the substation

The investment cost of a substation has consisted of the constant cost of substation construction and the investment cost of the transformer. The expression is shown as follows.

$$f_3 = a_{b0} + \sum_{i=1}^{N(t)} (a_b + b_b S_i) \tag{12}$$

where a_{b0} is the constant cost of substation construction, including the investment costs of the substation and other equipment. a_b is the constant cost of a single transformer, which mainly reflects the costs of land, incoming and outgoing intervals. b_b is the cost factor associated with the transformer capacity.

• Operation and maintenance costs of the substation

$$f_4 = op + mp \tag{13}$$

where op is the operating cost of the substation, which includes the load loss and the reactive power compensation devices required for renewable energy access, and is calculated as follows.

$$op = \sum_{i=1}^{N(t)} \omega \Delta P_d(i)(\frac{\pi l^2 \sigma(t)}{S_{total} \cos\varphi}) \tag{14}$$

where ω is the electricity price, $\Delta P_d(i)$ is the unloaded loss of the ith transformer, S_{total} is the total capacity of the substation, cosφ is the power factor of the transformer.

mp denotes the maintenance cost of the substation, which is generally taken as a percentage of the investment cost.

$$mp = b_1 f_1 + b_2 f_2 + b_3 f_3 \tag{15}$$

where b_1, b_2 and b_3 are the factors of cost (f_1), cost (f_2) and cost (f_3), respectively.

• Reliability Cost

The number of outgoing lines will be affected by the number of substation capacity units. Further, the combination of different numbers of outgoing lines and wiring methods will affect reliability costs, as shown in Table 1.

$$f_5 = \sum_{k=1}^{N(t)} (L_{Ak} + L_{Bk} + L_{Ck}) \times \omega \tag{16}$$

where L_{Ak}, L_{Bk}, L_{Ck} are the average outage losses in year k of partitions A, B, and C, respectively.

3.2. Constraints

Section headings should be left justified, bold, with the first letter capitalized and numbered consecutively, starting with the Introduction. Sub-section headings should be in capital and lower-case italic letters, numbered 1.1, 1.2, etc, and left justified, with second and subsequent lines indented. All headings should have a minimum of three text lines after them before a page or column break. Ensure the text area is not blank except for the last page.

• The constraint with the numbers of incoming and outgoing lines

Since the number of incoming and outgoing lines in the substation needs to comply with the relevant local distribution network planning guidelines, the number of lines for each wiring means on the 10 kV side needs to satisfy the following constraints.

$$M_1 \min \leq M_1(t) \leq M_1 \max \tag{17}$$

$$M_2 \min \leq M_2(t) \leq M_2 \max \tag{18}$$

$$N_1(t) + N_2(t) + N_3(t) = M_2(t)N(t) \tag{19}$$

where $N_1(t)$, $N_2(t)$, $N_3(t)$ are the numbers of wiring in multi-segment multi-contact network, single ring network, and double ring network, respectively.

It is worth noting that, based on the reliability targets for each type of power supply area and the reliability indicators for typical wiring patterns, the wiring methods and their corresponding numbers applicable to each type of power supply area can be roughly screened. The distribution network planned will necessarily meet the reliability requirements (lower limit) [16], but the specific combination of different network structures depends on the management target.

• "$N-1$" constraints on transformers and lines

"$N-1$" constraint of transformers means when a transformer is withdrawn from operation due to a fault, the remaining transformers bear the entire load without overloading. The constraint inequality of capacity and number of transformers is as follows.

$$\frac{\pi R^2\sigma(t)}{[N(t)-1]S_i\cos\varphi}\leq k_1 \tag{20}$$

where k_1 is the short-time overload rate of the transformer.

"$N-1$" constraint of lines means when a line is withdrawn from operation due to a fault or maintenance, the remaining lines can carry the full load without overloading. The constraint inequalities are as follows.

$$\frac{\pi R^2\sigma(t)}{[M_1(t)-1]\varDelta S_1\cos\varphi_1}\leq k_2 \tag{21}$$

$$\frac{\pi R^2\sigma(t)}{[M_2(t)-1]\varDelta S_2\cos\varphi_2}\leq k_3 \tag{22}$$

where $\varDelta S_1$ and $\varDelta S_2$ are the maximum capacity of the incoming and outgoing lines, respectively. k_2 and k_3 are the short-time allowable overload rates of incoming and outgoing lines, respectively. $\cos\varphi_1$ and $\cos\varphi_2$ are the power factors of incoming and outgoing lines, respectively.

The constraint of delivery capacity is as follows.

$$(N_1\times\beta_1\%+N_2\times\beta_2\%+N_3\times\beta_3\%)\times S_{line}\geq S_{total} \tag{23}$$

where β_1, β_2 and β_3 are the maximum load factor of lines 1, 2 and 3, respectively. S_{line} is the maximum transmission capacity of the line.

• Constraint on transformer capacity–load ratio

The capacity–load ratio E is the ratio of the transformer's capacity to its maximum load, and the following constraints should be satisfied in the transformer capacity planning.

$$E_{\min}\leq\frac{\sum_{i=1}^{N(t)}S_i\cos\varphi}{\pi R^2\sigma(t)}\leq E_{\max} \tag{24}$$

where $E_{\min}$ and $E_{\max}$ are the minimum and maximum values of the capacity–load ratio, respectively.

4. Algorithm

Substation planning is affected by dynamic changes in load density and renewable energy output, and there are more options for capacity and number of station transformers. Meanwhile, the model constructed in the previous section is characterized by many variables and complex constraints. The MVO algorithm is a meta-heuristic optimization algorithm proposed in 2016 with stronger optimality finding capability [17–19]. Therefore, combining the model characteristics and algorithm advantages, this paper adopts the MVO algorithm to solve the model.

The feasible solutions in the MVO algorithm correspond to universes, and the fitness of the solutions corresponds to the expansion rate of that universe. In each iteration, the universes are sorted according to the expansion rate, and a universe is randomly selected as a white hole through roulette, and the universes exchange matter through black and white holes. Assume that the initial value of the multi-universe population is :

$$\boldsymbol{Z}=\begin{bmatrix} z_1^1 & z_1^2 & \cdots & z_1^d \\ z_2^1 & z_2^2 & \cdots & z_2^d \\ \vdots & \vdots & & \vdots \\ z_s^1 & z_s^2 & \cdots & z_s^d \end{bmatrix} \tag{25}$$

where d is the number of parameters; s is the number of universes.

According to the roulette mechanism update, a white hole or black hole transfers cosmic objects body as shown in Eq. (26).

$$z_i^j=\begin{cases} z_k^j, r_1<\boldsymbol{N}_C\left(\boldsymbol{Z}_i\right) \\ z_i^j, r_1\geqslant\boldsymbol{N}_C\left(\boldsymbol{Z}_i\right) \end{cases} \tag{26}$$

where $\boldsymbol{Z}_i$ is the ith universe; $\boldsymbol{N}_C(\boldsymbol{Z}_i)$ is the normalized expansion rate of the ith universe, which differs for each individual universe; r_1 is the random value of [0,1]; z_k^j is the jth dimensional component of the kth universe selected by the roulette wheel.

Matter is randomly transferred between universes through wormholes to ensure population diversity, while all exchange matter with the optimal universe to increase the expansion rate.

$$z_i^j = \begin{cases} \{z_j + K_{\mathrm{TDR}}\left[\left(uc_j - wc_j\right) \times r_4 + wc_j\right], r_3 < 0.5 \\ z_j - K_{\mathrm{TDR}}\left[\left(uc_j - wc_j\right) \times r_4 + wc_j\right], r_3 \geqslant 0.5 \\ z_i^j, r_2 \geqslant K_{\mathrm{WEP}} \end{cases} \tag{27}$$

where K_{WEP}, K_{TDR} are the wormhole existence rate and travel distance, respectively. r_2, r_3, r_4 are random values in [0,1]. uc_j and wc_j are the upper and lower bounds of the dynamic boundary, respectively.

The wormhole existence rate K_{WEP} and the travel distance rate K_{TDR} are two extremely important parameters in the MVO algorithm that need to be increased linearly over iterations in order to emphasize the search during the optimization process. The variation of K_{WEP}, K_{TDR} in the iteration is able to get a more accurate global/local search in the best universe.

$$K_{\mathrm{WEP}} = K_{\mathrm{WEP}_{\min}} + t \times \left(\frac{K_{\mathrm{WEP}_{\max}} - K_{\mathrm{WEP}_{\min}}}{T}\right) \tag{28}$$

$$K_{\mathrm{TDR}} = 1 - \frac{t^{1/p}}{T^{1/p}} \tag{29}$$

where $K_{\mathrm{WEP}_{\max}}$ and $K_{\mathrm{WEP}_{\min}}$ are the upper and lower bounds of parameter K_{WEP}, which are taken as 1 and 0.2, respectively; t is the current number of iterations; T is the maximum number of iterations; p is the development accuracy of the algorithm, $p = 6$.

5. Simulation

The simulation in this Section is based on a computer with a CPU of Intel Core12700 and 16 GB of memory, and the model is programmed in Python and solved by MVO algorithm.

5.1. The setting of simulation parameters

Assuming that the regional load density is 10 MW/km2 and the regional power supply radius is 1.5 km. The annual growth rate of the load is assumed to be given. In order to simplify the dynamic changes of renewable energy growth and load growth, the renewable energy growth is equivalent to reducing the load density, reducing the annual load growth rate to 5%. The Operator and Maintainer coefficients b_1, b_2, and b_3 are the same, 0.032. The 10 kV side outgoing line adopts three types of wiring: multi-contact and multi-section network, single-loop network and double-loop network, and the wires model used are JKLYJ-240, YJV22-3 × 400 and YJV22-3 × 300 respectively, with the maximum allowable load capacity corresponding to 550 A, 550 A and 600 A. The number of feeder switches has been optimized according to the conclusions of the literature [20] to ensure a better reliability. The current 110 kV substation transformer capacity is mostly $3 * 50$ MVA, or $3 * 63$ MVA, and this optimization focuses on the preferred high-capacity substation, of which the single transformer capacity is 63 MVA or 80 MVA and the number of transformers is 3 or 4. The remaining parameters of the simulation are shown in Table 2.

Table 2. The simulation parameters of transformer with capacity of 63 MVA and 80 MVA.

Parameters	63 MVA	80 MVA
Electricity price ω (yuan)	0.6	0.6
Transformer price (million yuan)	14.3	16
the constant cost of each incoming line C_0 (million yuan)	0.8	1
investment cost of a unit length of the incoming line C_1 (million yuan/km)	3	3.5
the constant cost of each outgoing line C_2 (million yuan)	0.8	0.8
investment cost of a unit length of out-line C_3 (million yuan/km)	0.95	0.95

5.2. Analysis of simulation results

This example compares the planning of high-capacity substations by considering renewable energy access or not. The simulation results are shown in Tables 3 and 4 below.

Table 3. The optimization results for large capacity substation (63 MVA).

Unit capacity	63 MVA (Renewable energy)			63 MVA (Without renewable energy)		
Original unit no.	3			3		
final scale	4			4		
Year of commissioning of new transformer	4th			3rd		
Year order	multi-segment multi-contact network	single ring network	double ring network	multi-segment multi-contact network	single ring network	double ring network
1	5	3	3	4	4	7
2	5	5	5	4	4	7
3	5	5	5	4	5	7
4	7	7	6	7	6	7
5	7	7	6	7	6	7
6	7	7	6	7	6	7
7	7	7	6	7	6	7
8	7	7	6	7	6	7
9	7	7	7	7	7	7
10	7	7	7	7	7	7
Reliability cost	21.43 million ¥			27.46 million ¥		
Total costs	1444.6489 million ¥			1452.3974 million ¥		

Table 4. The optimization results for large capacity substation (80 MVA)

Unit capacity	80 MVA (Renewable energy)			80 MVA (Without renewable energy)		
Original unit no.	2			2		
final scale	3			3		
Year of commissioning of new transformer	3rd			3rd		
Year order	multi-segment multi-contact network	single ring network	double ring network	multi-segment multi-contact network	single ring network	double ring network
1	4	4	3	5	5	4
2	4	7	6	5	7	5
3	4	7	7	7	7	6
4	4	7	7	7	7	6
5	4	7	7	7	7	6
6	5	7	7	7	7	7
7	7	7	7	7	7	7
8	7	7	7	7	7	7
9	7	7	7	7	7	7
10	7	7	7	7	7	7
Reliability cost	22.80 million ¥			25.64 million ¥		
Total costs	1412.4109 million ¥			1414.9560 million ¥		

It can be seen from Table 3 that the number of transformers put into the substation in the first year is 3 when the transformer capacity is 63 MVA. When considering renewable energy access, the fourth transformer was put into the substation in the fourth year. The initial number of outgoing lines is as follows: 5 for multi-segment multi-contact network, 3 for single ring network, and 3 for double ring network, as a total of 11. The final number of lines for all three wiring types is 7. The total cost of this scheme is 1444.6489 million ¥. When renewable energy access is not considered, the fourth transformer was put into the substation in the third year. The initial number of outgoing lines is as follows: 4 for multi-segment multi-contact network, 4 for single ring network, and 7 for double ring network, as a total of 15. The final number of lines for all three wiring types is 7. The total cost of this scheme is 1452.3974 million ¥.

When the capacity of the transformer is selected as 80 MVA, 2 transformers were put into use for the substation in the first year. When considering renewable energy access, the third transformer was introduced into the substation in the third year. The similar large capacity substation, as compared to the previous scheme $4*63$ MVA, eventually forms a $3*80$ MVA structure. The initial number of outgoing lines is as follows: 4 for multi-segment multi-contact network, 4 for single ring network, and 3 for double rings network. The final number of lines for all three wiring structures is 7. The total cost of this scheme is 1412.4109 million ¥. When renewable energy access is not considered, the third transformer was put into the substation in the third year. The initial number of outgoing lines is as follows: 5 for multi-segment multi-contact network, 5 for single ring network, and 4 for double ring network, as a total of 14. The final number of lines for all three wiring types is 7. The total cost of this scheme is 1414.9560 million ¥.

The addition of renewable energy sources allows the reliability of the area covered by the large capacity substation to be increased, which can reduce reliability costs. At the same time, the renewable energy source can offset some of the load growth and delay the commissioning of the new main transformer resulting in lower total investment costs. It can be seen that when a 63 MVA transformer is selected as the main transformer for a large capacity substation, considering the access of renewable energy can reduce the substation investment by approximately ¥7.7 million, and when an 80 MVA transformer is selected as the main transformer for a large capacity substation, it can reduce the substation investment by approximately ¥2.5 million.

By comparison of the examples, the total cost of the $3*80$ MVA scheme is lower, so the distribution companies in the area prefer to build $3*80$ MVA large capacity substations based on this method.

6. Conclusion

Considering the variation of renewable energy penetration rate and load, this paper proposes a method to optimize the capacity of substations in distribution networks. The economic analysis model for optimizing the capacity and the number of a substation transformer in the distribution network considers the renewable energy access and the reliability of the secondary network of the substation. Various classical wiring structures of the distribution network are retained and the impact of renewable energy access on the reliability of the distribution supply is analyzed and quantified. Moreover, reliability indicators are transformed into economic indicators in the form of energy outage losses. The optimization model is solved by the MVO algorithm with satisfactory results in test examples, which are employed to demonstrate and verify the validity and rationality of the method.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

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